Project Report: Reconstructing Historical F1 Standings Using XGBoost-based Soft Scaling and Transformer-LSTM Hybrid Models

1. Abstract

This project aims to reconstruct historical Formula 1 standings(1950-2024) by simulating modern “Sprint” race formats using a deep learning approach. The primary challenge in analyzing historical F1 data is the lack of temporal context and the high noise levels inherent in raw race results. To address this, develop a hybrid model combining Long Short-Term Memory(LSTM) networks with an Attention Mechanism. Furthermore, to overcome the “black box” initialization problem of deep learning, we introduced a novel “Soft Scaling” technique using XGBoost. This method extract feature importance to initialize the neural network’s weights. The proposed approach achieved a 13.3% reduction in Mean Absolute Error (MAE) through feature engineering and an additional 8.4% improvement through XGBoost-based tuning. The final simulation suggests significant versions to historical World Driver’s Championship titles, highlighting a more meritocratic distribution of championships based on pure performance data.

2. Introduction

Formula 1 is sport defined by continuous evolution, making historical comparisons difficult. Traditional statistical models often fail to capture the “momentum” of driver or the “form” of a constructor because they treat race results as isolated events rather than continuous time series flow.

The objective of this project is two-fold, first Predictability, build a model capable of accurately predicting race outcomes specifically sprint formats by learning from sequential historical data. Next, reconstruction, apply this model to entirety of F1 history to see how modern scoring and performance metrics would alter historical championship outcomes.

We hypothesize that by combining the sequential memory of LSTMs with the selective focus of Attention mechanisms, and seeding the model with domain knowledge derived from XGBoost, we can significantly outperform baseline predictions.

3. Dataset

The dataset encompasses the entire history of Formula 1 World championship from 1950 to 2024

Data collection and consolidation, raw data was aggregated from multiple CSV sources, including circuits.csv, dirvers.csv, results.csv, and races.csv and more. These were merged using unified keys like raceId, driverId, constructorId to create a structural baseline. Feature Engineering, a critical finding of this project was that raw race records like Grid, Points, Laps failed to capture temporal context. To address this, engineered a v2 dataset focused on relative pace, which replacing static qualifying rankings with the time gap to the pole position, quantifying absolute vehicle competitiveness. Contextual signals, creating driver_Fform and team_form feature to capture momentum over recent races. Track Proficiency, circuit_avg_pos was introduced to quantify a driver's historical performance on specific track layouts.

4. Models

The core innovation of this project lies in the Hybrid Architecture and the XGBoost Soft Scaling initialization strategy.

Architecture is LSTM + Attention. LSTM, we utilized a 2-layer LSTM with 64 hidden units to capture the sequential dependency of race results. This allows the model to remember long-term trends such as car upgrades or driver confidence. Attention mechanism, a mechanism was added to dynamically weight historical data. It assigns higher relevance to recent races last 3 rounds and performance on similar tracks, filtering out historical noise.

Hyperparameter Tuning via XGBoost Insights. To optimize the neural network, we moved away from random weight initialization. We use XGBoost to perform a quantitative feature attribution analysis. Extraction, XGBoost identified non-linear feature relationships and assigned importance scores. Transfer, these importance scores were mapped directly into the Priority Layer of deep learning model as initial weights. Optimization, this allowed the model to start training from an informed baseline rather than random noise, ensuring faster and more precise convergence.

5. Experiment and Result

Experimental setup as follows, optimizer use Adam with learning rate 0.001, loss function with Mean Squared Error, Epochs with 100 and Sequence length with 3 predicting the next race based on the previous 3. Quantitative analysis, we conducted controlled experiments to measure the impact of both feature engineering and model tuning. Using the identical LSTM architecture, the engineered dataset showed drastic improvements over raw data. MAE reduced from 3.46 to 3.00 and MSE reduced from 17.57 to 13.96. Impact of XGBoost Soft Scaling comparing the Base Hybrid model against the Tuned Hybrid XGBoost initialized model revealed further gains. MAE reduced from 3.0038 to 2.7526 and MSE reduced from 13.9592 to 11.8846. Simulation result with historical reconstruction. Applying the optimized model to reconstruct the World Driver's Championship history resulted in notable changes to the record books. Most titles Lewis Hamilton with 8 times, Michael Schumacher with 7times, and Alain Prost with 7times. The model suggested that based on pure performance metrics excluding mechanical failures or bad luck, Alain Prost would have secured 7 titles historically 4, and Fernando Alonso would have won the 2007 championship.

6. Conclusion

This project successfully demonstrated that integrating domain-specific feature engineering with a hybrid LSTM-Attention architecture significantly enhances the prediction of F1 race outcomes. The introduction of XGBoost-based Soft Scaling proved to be a highly effective method for hyperparameter tuning, acting as a bridge between interpretable tree-based models and complex deep learning networks. The resulting MAE of 2.75 establishes a robust baseline for sports analytics, proving that AI can effectively quantify performance momentum in chaotic environments.